

Prop. If $p > 0$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \sqrt[n]{p} = 1.$$

Proof. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $N > \frac{1}{p\varepsilon}$. Then

$$n \geq N \Rightarrow \frac{1}{n^p} \leq \frac{1}{N^p} < \varepsilon$$

And, if $p = 1$, directly $\sqrt[n]{p} = 1 \rightarrow 1$.

If $p > 1$, let $x_n = \sqrt[n]{p} - 1$. Then, $x_n > 0$, and

$$p = (x_n + 1)^n \geq nx_n \Rightarrow x_n \leq \frac{p}{n}, \text{ so } x_n \rightarrow 0 \text{ as } n \rightarrow \infty.$$

If $p < 1$, then $\sqrt[n]{p} \rightarrow 1$, so $\sqrt[n]{p} \rightarrow 1$.

Prop. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.

Proof. Let $x_n = \sqrt[n]{n} - 1$.